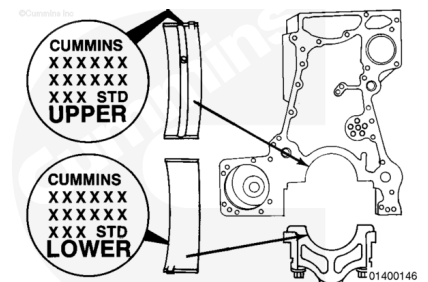


Trainings ▾

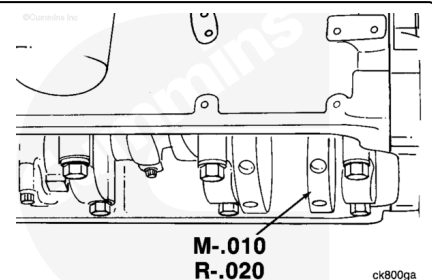
General Information

The upper bearings contain an oil hole. The lower bearings do **not** have an oil hole. Both bearings are marked on the backside indicating location (upper or lower) and size (standard [STD] or oversize [OS]). The amount of the bearing is oversized is indicated in U.S. customary inches.

Use the same size bearing standard, 0.254 mm [0.010 in], 0.508 mm [0.020 in], or 0.762 mm [0.030 in] that was removed.

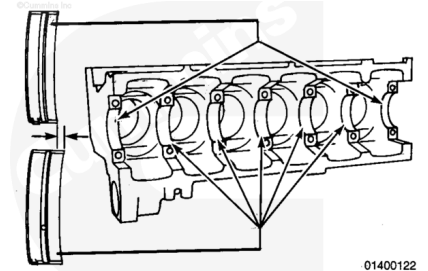


The crankshaft will be stamped on the end of the number one counterweight to indicate if it has been ground undersize. Thrust bearing size is stamped on a crankshaft counterweight, adjacent to the thrust location.



The location number of the main bearing caps, beginning at the front, are numbers 1 through 7.

The main bearings are two widths. The **narrow** main bearings fit locations number 1 and number 7. The **wide** bearings fit the remaining locations.



Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged

contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

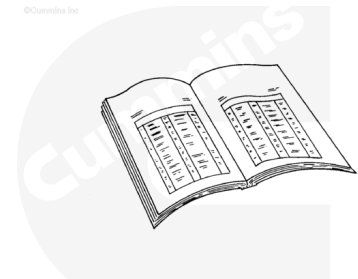
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>) .
- Remove the oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)
- Remove the lubricating oil pan adapter cover. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html>)
- Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-035.html>)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-027-tr.html>)



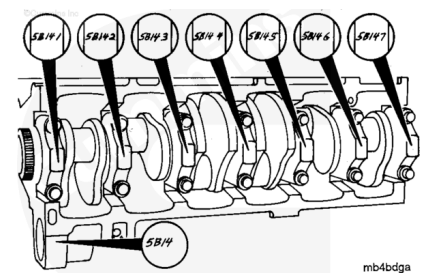
ck800wa

Remove

⚠ WARNING ⚠

Replace the main bearings one at a time. Personal injury and damage to the crankshaft can result if the crankshaft falls.

⚠ CAUTION ⚠



mb4bdga

Do not rotate the crankshaft until all of the main bearing caps are pulled to the block. Engine damage will result if the bearings move out of location.

The main bearing caps **must** be pulled into position against the block before the crankshaft is rotated. As main bearing caps are replaced, tighten the cap to a minimum of 68 N•m [50 ft-lb] before turning the crankshaft.

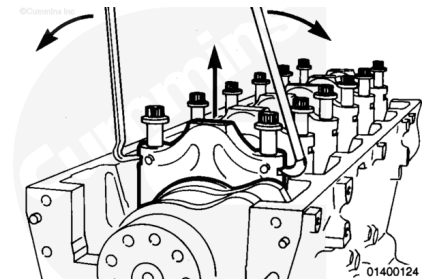
The main bearing caps **must** be marked for position. The last number on each bearing cap identifies the cap location position in the block.

The block and main bearing cap identification numbers **must** be identical.

Use a steel stamp and mark any cap that is **not** marked correctly.

Loosen the main bearing cap capscrews until there is approximately 13 mm [0.5 inch] between the capscrew head and the cap. Use two pry bars to loosen the cap. Install the pry bars in the notches at the sides of the caps.

Remove the cap and capscrews.



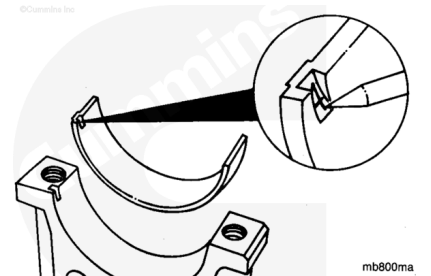
Mark the bearing position for future identification or for possible failure analysis.

Remove the lower bearing. Use an awl to mark the bearing position in the tang area.

Refer to Procedure 001-007 in Section 7.

(/qs3/pubsys2/xml/en/procedures/20/20-001-007-tr.html)

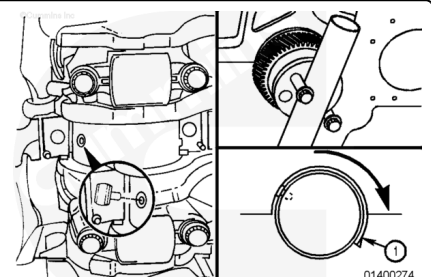
Remove the crankshaft thrust bearings.



To remove the upper main bearing shell, use the main bearing roll-out tool, Part Number 3823818, and install it in the crankshaft main bearing journal oil hole.

Rotate the crankshaft so the tang (1) of the main bearing shell rolls out of the block first.

Use the crankshaft adapter mounting bolts and slowly turn the crankshaft until the bearing shell is out of the block.



Make sure the bearings are marked for location. The bearings **must** be installed in the original location after rebuild.

Clean and Inspect for Reuse

⚠ WARNING ⚠

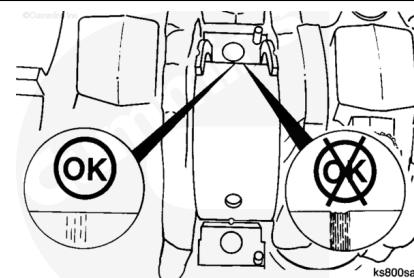
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

To reduce the possibility of bearing damage, do not use a scraper or wire brush on bearing surfaces.



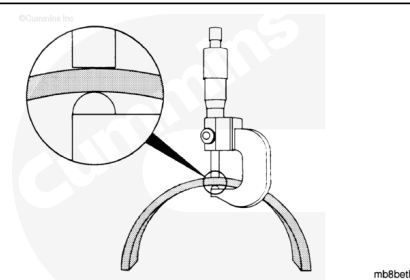
Make sure the bearings are marked for location. The bearings **must** be installed in the original location after rebuild.

Use a lint-free cloth. Clean the crankshaft journals, and connecting rods, and main bearings. Check the crankshaft journal for damage.

Use solvent to clean the bearings.

Dry with compressed air.

Use ball-end micrometer. Measure the bearing thickness at the wear location. The bearing **must** be replaced if it is **not** within specifications.



Main Bearing Thickness Standard or Oversize

	mm		in
Standard	4.280	MIN	0.1685
	4.348	MAX	0.1712

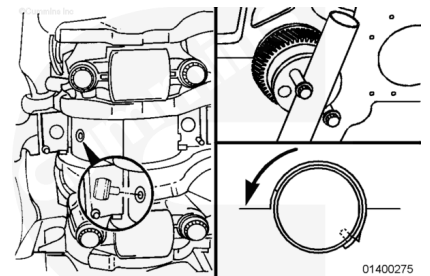
Oversize 0.010 mm	4.407	MIN	0.1735
	4.475	MAX	0.1762
Oversize 0.020 mm	4.534	MIN	0.1785
	4.602	MAX	0.1812
Oversize 0.030 mm	4.661	MIN	0.1835
	4.729	MAX	0.1862
Oversize 0.040 mm	4.788	MIN	0.1885
	4.844	MAX	0.1907

Install

Reused bearings **must** be installed in their original location.

Install the upper main bearing shells using the same method that was used for the removal of the shells.

The bearing tang (1) **must** fit into the slot (2) in the bearing saddle to secure the bearing in the proper location.



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Before installing the thrust bearings, Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-006-tr.html)

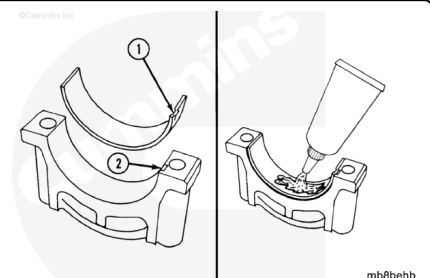
Use a lint-free cloth. Clean the lower main bearings, the lower thrust bearings, and the mounting surfaces.

Do **not** lubricate the back of the main bearings.

Align the tang (1) in the bearing with the slot (2) in the main bearing cap. Install the bearing. The end of the bearing **must** be even with the main bearing cap mounting surface.

Lubricate the bearing surface with clean engine oil.

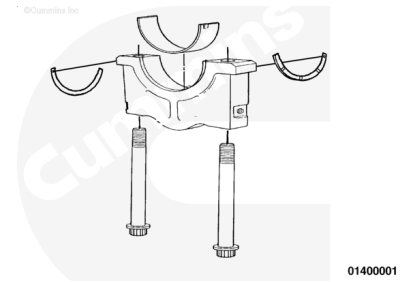
Install the upper thrust bearings. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-007-tr.html)



mb8behb

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the grooves in the thrust bearings must point toward the crankshaft. The dowels that secure the bearings must not protrude above the bearing.



Lubricate the capscrew head and threads with SAE EP 140W oil. Allow the excess oil to drip off of the capscrews before installing in the block.

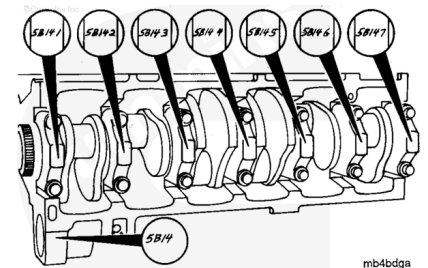
Install the capscrews in the cap.

Install the two thrust bearings on the number six main bearing cap. Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-007-tr.html)

⚠ CAUTION ⚠

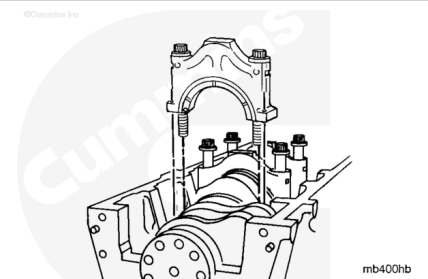
To reduce the possibility of engine damage, the numbers on the main bearing caps must be the same as the numbers on the block.



Check the number on the main bearing caps. The last digit of each number (1 through 7), indicates the correct location.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the side of the cap with the bearing tang slot and the bearing locking tang are toward the same side of the block.

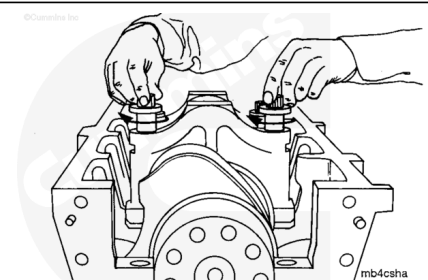


Lubricate the bearing surface with engine oil. The bearing shell **must** be firmly seated in the cap and the correct capscrew, washer combination in position. Install the main bearing caps.

Do **not** hit the main bearing caps with a hammer. The bearing shells can fall out.

⚠ CAUTION ⚠

Do not rotate the crankshaft until all of the main bearing caps are pulled to the block. Damage will result if the bearings move out of location.



Use your hands to install the capscrews.

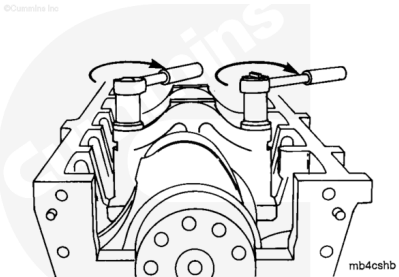
Turn each capscrew until it touches the main bearing caps.

Note : Do not use an impact wrench. The main bearing shells can fall out.

Use both of the capscrews to pull the main bearing cap into position.

Use two wrenches and tighten both capscrews at the same time.

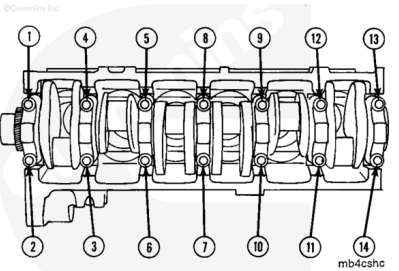
Check to be sure the number 7 cap is even with the back of the block.



Use the following steps and tighten the capscrews in the sequence shown.

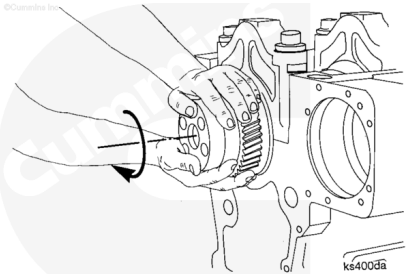
Torque Value:

1. 265 n•m [195 ft-lb]
2. 605 n•m [445 ft-lb]
3. Loosen all.
4. 265 n•m [195 ft-lb]
5. 605 n•m [445 ft-lb]



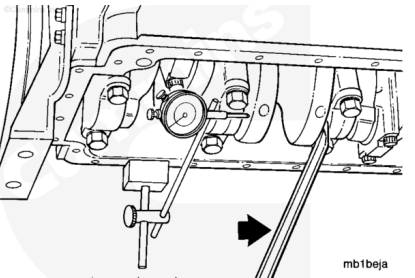
Turn the crankshaft. If the pistons have been removed, use your hands. If the pistons have **not** been removed, use the barring mechanism.

Note : The crankshaft will turn freely if the main bearings are installed correctly.



Use a dial indicator and measure the crankshaft end clearance.

Crankshaft End Clearance		
mm		in
0.10	MIN	0.004
0.41	MAX	0.016



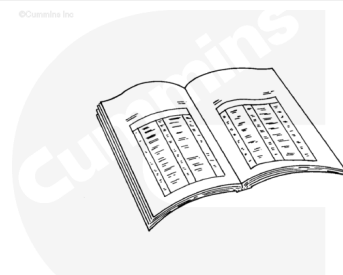
If the clearance is **not** within specifications, check for foreign material. Oversize thrust bearings are available to adjust the end clearance.

Note : The upper and lower thrust bearing on the same side of the main caps must be the same thickness.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-027-tr.html>)
- Install the oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-035.html>)
- Install the oil pan adapter cover. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-026-tr.html>)
- Install the oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Operate the engine. Check for leaks.